Department of Engineering Science
School of Science & Technology



CES 440 Data Communications
Ch.1 Introduction

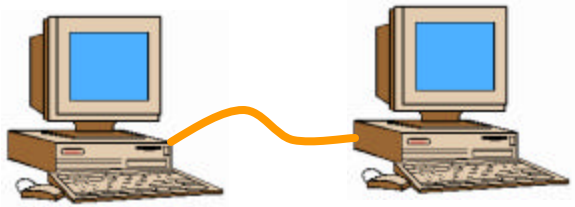
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OUTLINE

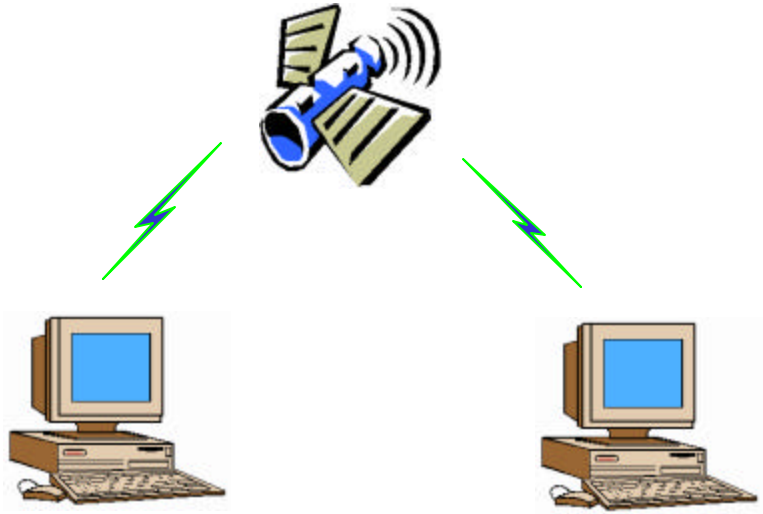
- **Computer Network Definition**
- **Applications of Computer Network**
- **Network Hardware & Classifications**
- **Network Software**
- **Reference Models**
- **Example Networks**
- **Network Standardizations**

DEFINITION: WHAT IS COMPUTER NETWORK?

- **Computer Network: a collection of autonomous computers interconnected with each other.**
  - Autonomous: independent, self-sufficient, self-ruling.
  - Interconnection: capable of exchanging information with one another.
    - Via copper wire, fiber optics, microwaves, infrared, etc.
- **Examples:**



Network 1



Network 2

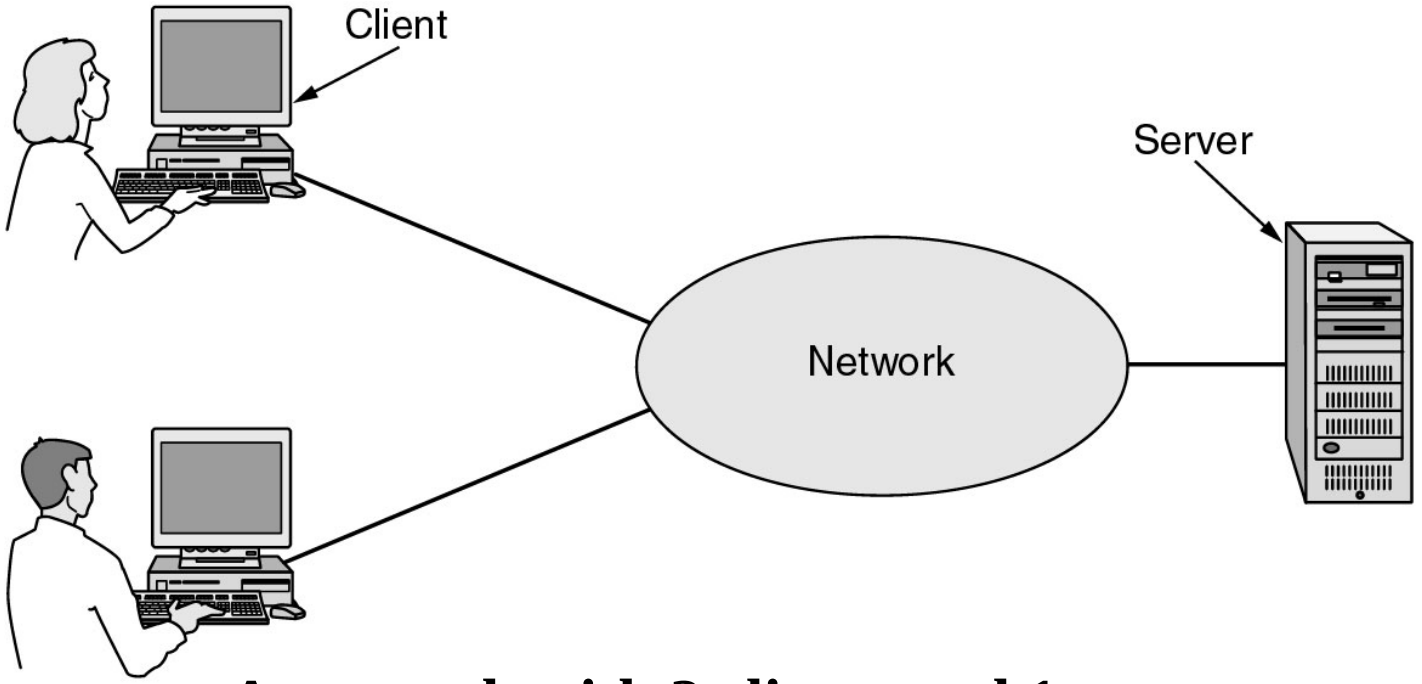
DEFINITION

- **Internet is not a computer network!**
  - Internet is a collection of computer networks
    - network of networks

- **World Wide Web (WWW) is not a computer network!**
  - WWW is an application that runs on top of Internet.

DEFINITION: NETWORK MODEL I

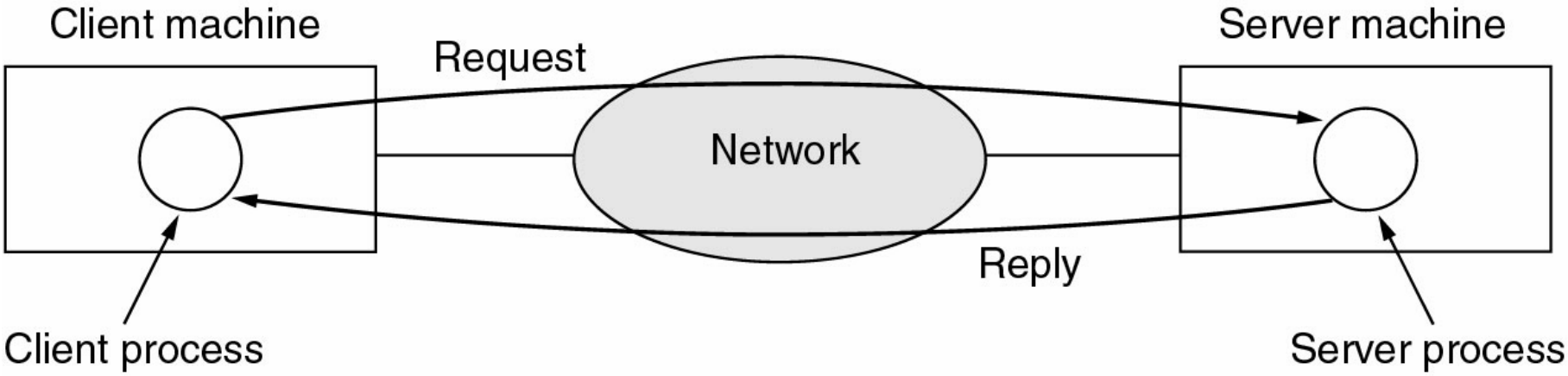
- **Client-Server Model**
  - Example: WWW is an application built on the client server model.



A network with 2 clients and 1 server.

DEFINITION: NETWORK MODEL I

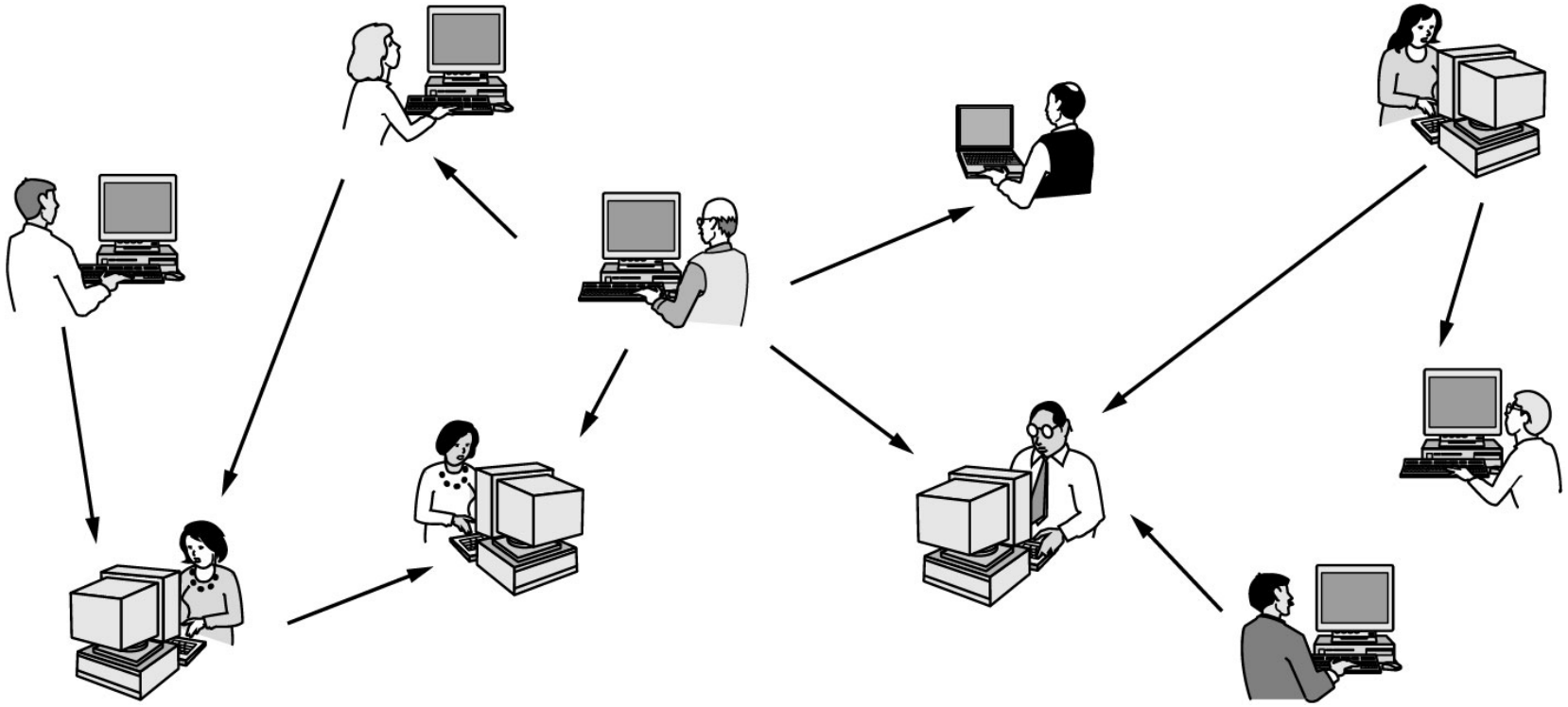
- **Client-Server Model (Cont'd)**
  - The client-server model involves Request and Reply



  - A computer can be both a client and a server at the same time
    - Client & Server are usually referring to the software running in the computer.
    - Client and server can be installed in one computer.

DEFINITION: NETWORK MODEL II

- **Peer-to-Peer Model (P2P)**
  - No fixed clients and servers.
  - Example: Napster



DEFINITION: METRIC CONVENTIONS

- **Metric conventions**
  - When measuring transmission speed or bandwidth
    - Kilo: 1000, Mega: 10^6 , Giga: 10^9
    - 1 Kbps: 1000 bits per seconds
    - 1 Mbps: 1×10^6 bits per seconds
    - 1 Gbps: 1×10^9 bits per seconds
  - When measuring storage size (memory, disk, file, etc.)
    - Kilo: 2^{10} , Mega: 2^{20} , Giga: 2^{30}
    - 1KB: 1×2^{10} bytes = $1\times2^{10}\times8$ bits
    - 1MB: 1×2^{20} bytes = $1\times2^{20}\times8$ bits
    - 1GB: 1×2^{30} bytes = $1\times2^{30}\times8$ bits

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- **Applications of Computer Network**
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APPLICATIONS: BUSINESS

- **Resource Sharing**
  - Printer, scanner, information, etc.
- **Communication Medium**
  - e-mail, video conference, etc.
- **E-Business, E-Commerce**

Tag	Full name	Example
B2C	Business-to-consumer	Ordering books on-line
B2B	Business-to-business	Car manufacturer ordering tires from supplier
G2C	Government-to-consumer	Government distributing tax forms electronically
C2C	Consumer-to-consumer	Auctioning second-hand products on-line
P2P	Peer-to-peer	File sharing

APPLICATIONS: HOME

- **Access to remote information**
  - Web Surfing, Online Library, News Group, Bulletin Board (BBS), etc.
- **Person to Person Communication**
  - E-mail, Instant Messaging (MSN), Internet Phone (Vonage), etc.
- **Interactive Entertainment**
  - Video on Demand, Online Gaming, etc.
- **E-Commerce**
  - Online Shopping (Amazon, eBay), Online Banking, Online Bill Paying, etc.

APPLICATIONS: MOBILE USERS

- **Portable Office**
  - Access email, files, faxes anywhere on the road.
- **Taxi/Truck Dispatcher, Military, Utility Metering, Inventory, etc.**
- **Wireless V.S. Mobile**
  - Fixed Wireless, Mobile Wireless

Wireless	Mobile	Applications
No	No	Desktop computers in offices
No	Yes	A notebook computer used in a hotel room
Yes	No	Networks in older, unwired buildings
Yes	Yes	Portable office; PDA for store inventory

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CLAFFICATION BY TX TECHNOLOGIES

- **Classification by Tx Technologies**
  - Broadcast links
  - Point-to-point links (unicast links)
- **Broadcast Links**
  - A single communication channel shared by all the machines.
  - Broadcast link ≠ Broadcasting
  - Broadcasting
    - Packet: short messages Tx in the network.
    - Address field within packet specifies intended recipient.
    - Broadcasting: addressing a packet to all machines in network.
    - Multicasting: addressing a packet to a subset of the machines.
    - Broadcasting can be achieved with point-to-point link (set the address field of the packet to all users in network).
  - Broadcast Links are usually used in small scale networks.

CLASSIFICATION BY TX TECHNOLOGIES

- **Point-to-Point Links**
  - Networks consisting of many connections between individual pairs of computers.
  - Point-to-point link $\neq$ unicasting
  - Unicasting: One Tx and One Rx (Receiver)
    - You can send a message to a specified user through broadcast link by specify the address of the user in packet.
    - All other users will simply discard the message if the destination is not for me.
  - Larger networks are usually point-to-point.
- **Point-to-point link is different from peer-to-peer communication model**
  - Point-to-point link describes the physical Tx technology employed in the network.
  - Peer-to-peer model represents the information exchanging method adopted by computers.
    - P2P can be employed in network with Broadcast Links. (Packet addressed to a specified user, no server)

CLASSIFICATION BY NETWORK SCALE

- **Classification By Network Scales**
  - Personal Area Network (PAN)
  - Home Network
  - Local Area Network (LAN)
  - Metropolitan Aera Network (MAN)
  - Wide Area Network (WAN)
  - Internetwork

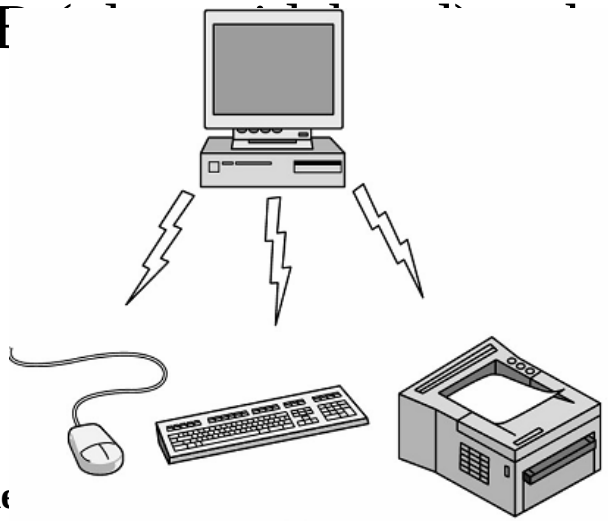
CLASSIFICATION BY NETWORK SCALE

- **Classification based on Network Scales (Cont'd)**

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	Local area network
100 m	Building	
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

CLAFFICATION : PAN

- **Personal Area Network (PAN)**
  - Networks that are meant for one person.
  - Could be either wired or wireless.
  - Usually all the network equipment are within a small space (several square meters).
  - Example: A PC with mouse, keyboard, and a printer.
  - Example: IEEE 802.15 (Wireless Personal Area Network with UWB [illegible] ology)



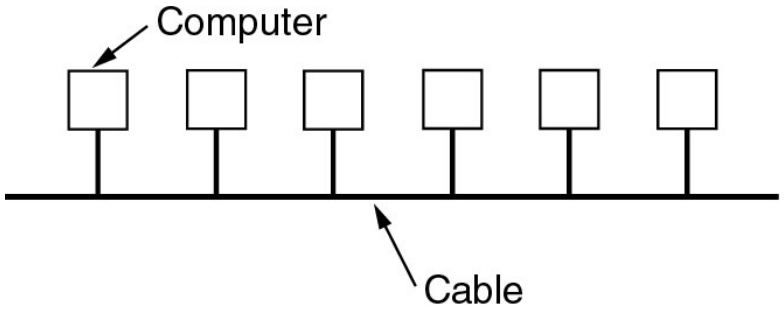
(a)

CLAFFICATION : LAN

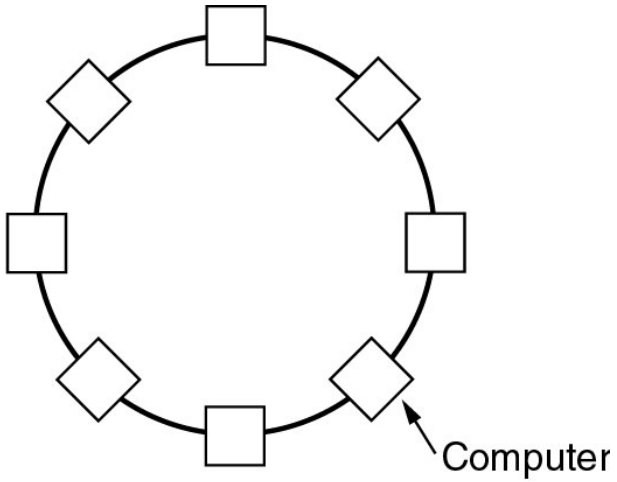
- **Local Area Network (LAN)**
  - Privately-owned network within a single building or campus
  - Up to a few kilometers in size.
  - Generally a broadcast network
- **Distinguished from other networks by three characteristics**
  - Restricted in size: worst case Tx time is bounded.
  - Transmission medium shared by all computers: Broadcast Links.
  - Unique Topologies (geometric layout)

CLAFFICATION : LAN

- **LAN Topologies**
  - Bus (a linear cable): IEEE 802.3 (Ethernet, most commonly used)
  - Ring: IEEE 802.5
  - Need a mechanism to resolve conflicts when two or more computers Tx simultaneously.



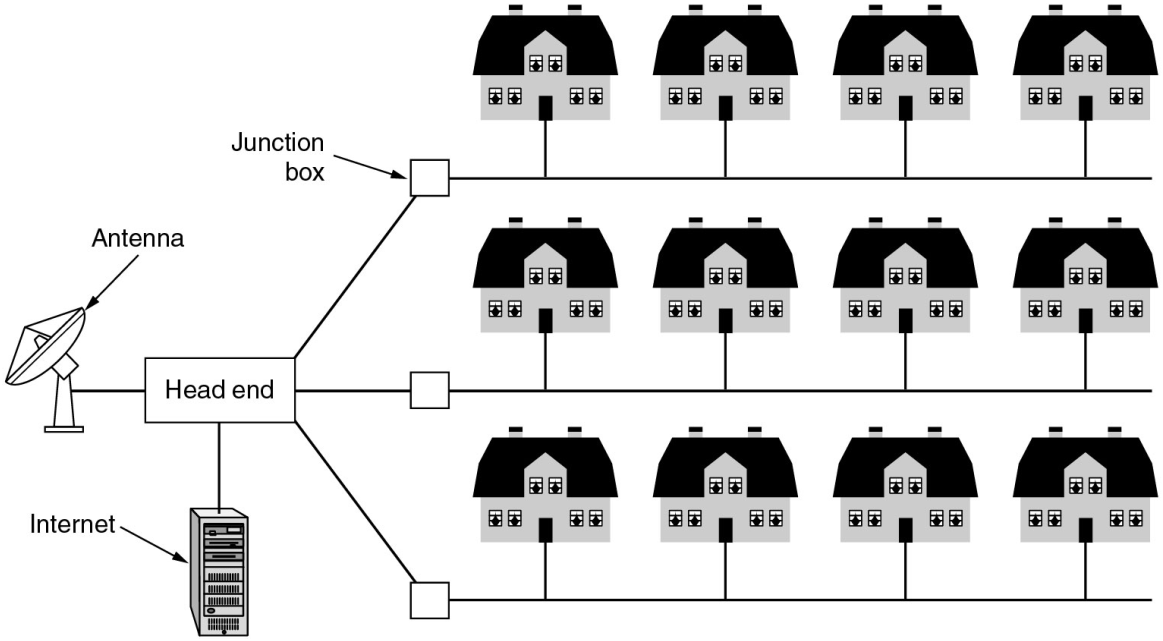
(a)



(b)

CLAFFICATION : MAN

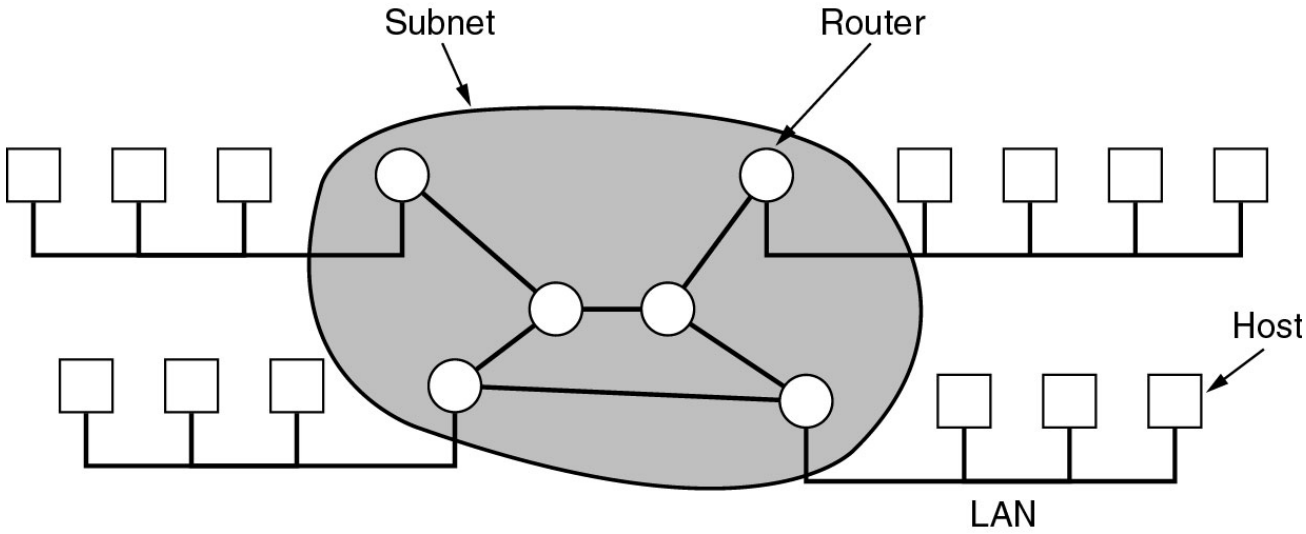
- **Metropolitan Area Network (MAN)**
  - The network covers a city.
  - Example: Cable TV Network (Hybrid Fiber Coax Network)



  - Example: WiMax (IEEE 802.16): wireless broadband access as an alternative to cable and DSL

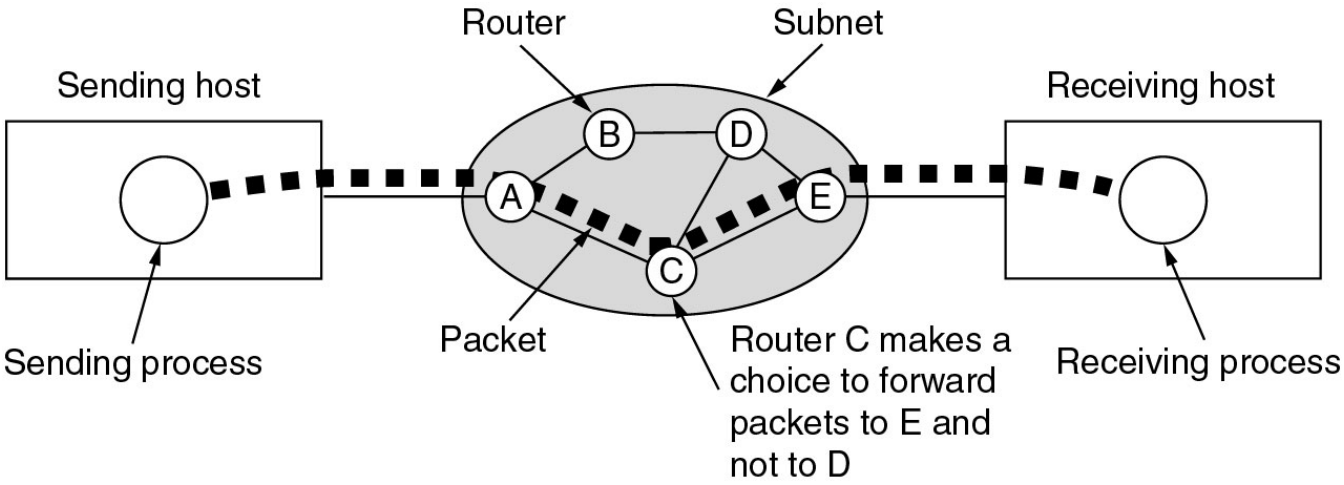
CLAFFICATION : WAN

- **Wide Area Network (WAN)**
  - Spans a large geographical area (country or continent)
  - Consists hosts (owned by customer) and subnets (owned by operator).
  - Subnet: collection of Tx Lines and switching elements to connect hosts.
    - Major function: transmit information from one point to another point.



CLAFFICATION : WAN

- **Switching Element (Router)**
  - Specialized computer connecting 2 or more Tx Lines.
- **Packet Switch (store-and-forward)**
  - Subnet moves packets from source to destination.
  - Packet is received at each intermediate router in its entirety, stored there, and forward to the next router.
  - Packet may use different route.
  - Router use routing algorithm to route packets.

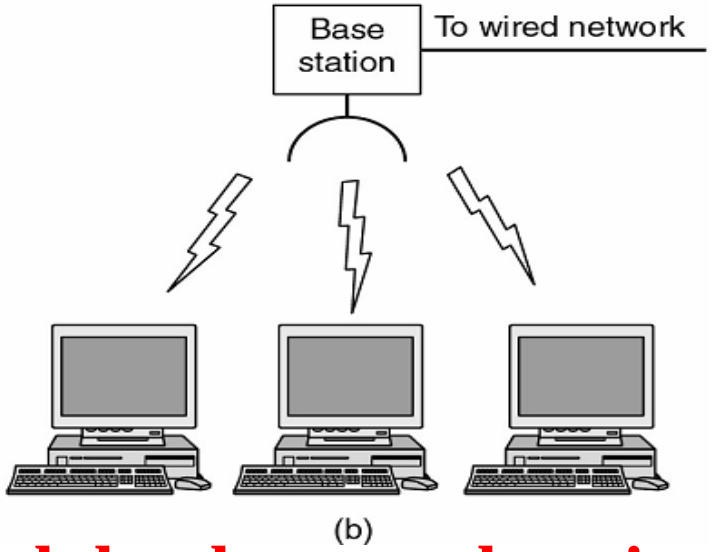


CLAFFICATION : WIRELESS NETWORK

- **Wireless Network Classifications**
  - System Interconnection
  - Wireless LAN
  - Wireless WAN
- **System Interconnection**
  - Interconnecting the components of a computer using short range radio.
  - Example: Bluetooth allows computer peripherals (digital cameras, printers, mouse, cell phone, etc.) to connect to a computer without cable. (Narrow Band)
  - Example: IEEE 802.16 (Broadband!)

CLAFFICATION : WIRELESS NETWORK

- **Wireless LAN**
  - Computers talk with base station via wireless radio.
  - Computer talk with computer (P2P) via wireless radio.
  - Standard: IEEE 802.11
- **Wireless WAN**
  - Cell Phone System.



(b)

- **Almost all Wireless network hook up to the wired network at some point.**

CLAFFICATION: HOME NETWORK

- **Home Network Categories**
  - Computers
    - Desktop, Notebook, PDA, etc.
  - Entertainment
    - TV, DVD, VCA, Stereo, Camcoder, etc.
  - Telecommunication
    - Phone, Cell Phone, Fax, etc.
  - Appliances
    - microwave, refrigerator, furnace, clock, AC, etc.
  - Telemetry
    - Utility meter, thermostat, smoke/burglar alarm, etc.
- **Objective: every device capable of communication with every other device, either wired or wireless.**

CLASSIFICATION: HOME NETWORK

- **Unique Properties**
  - Interconnectivity
    - Different Hardwares and Software
  - Easy to install and operate.
  - Low cost.
  - Security!
- **Example: ZigBee**
  - an association of companies working together to instill everyday devices with the ability of reliable, cost-effective, low-power, wireless communication.

CLASSIFICATION: INTERNETWORKS

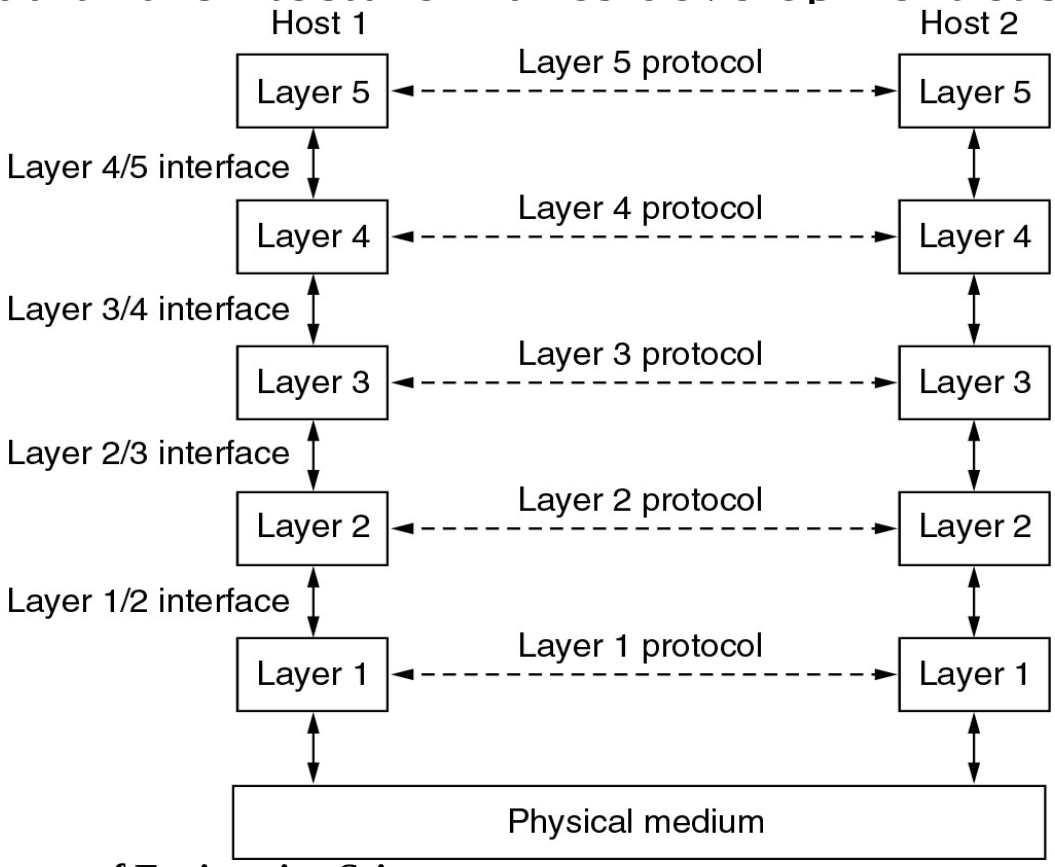
- **Definition**
  - A collection of interconnected networks (usually incompatible with each other) is called an internetwork or internet.
- **An internetwork is formed when distinct networks are interconnected**
  - Eg. LANs connected by subnet.
- **Internet V.S. internet**
  - Internet is an example of internet.

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SOFTWARE: PROTOCOL HIERARCHIES

- **To reduce complexity, network software is organized as a stack of layers.**
  - Modular architecture makes development easier and cheaper.



SOFTWARE: PROTOCOL HIERARCHIES (VERTICAL)

- **Each layer provides services to the layers above**
  - Via the interface between layers.
  - Details of the implementation of each layer is hidden to the higher layers → Data Encapsulation.
- **Between each adjacent pair of layers is an interface**
  - Interface clearly defines the services the lower layer makes available to the upper layer.
  - Clear cut interface make it simpler to change layer implementation → different host can use different implementation.
- **A layer in a machine is represented by an entity in that machine.**
  - Entity (SW or HW) does the function of that layer.

SOFTWARE: PROTOCOL HIERARCHIES (HORIZONTAL)

- **Peer Entities**
  - Entities in the same layer, but in different machines.
  - Layer *n* on one machine carries on conversation (exchanges information) with layer *n* on another machine.
  - Peer entities communicate with each other using protocol.
- **Protocol: An agreement between communication parties on how communication is to proceed.**
  - One of the most important concepts in computer network!
  - Each layer has its own protocol.
  - The protocol of a certain layer can only be understood by its peer entities.
- **Physically, peer entities do not talk with each other directly.**
  - Using the layers below them until the physical medium.
  - Actual communication occurs through physical medium.

SOFTWARE: PROTOCOL HIERARCHIES

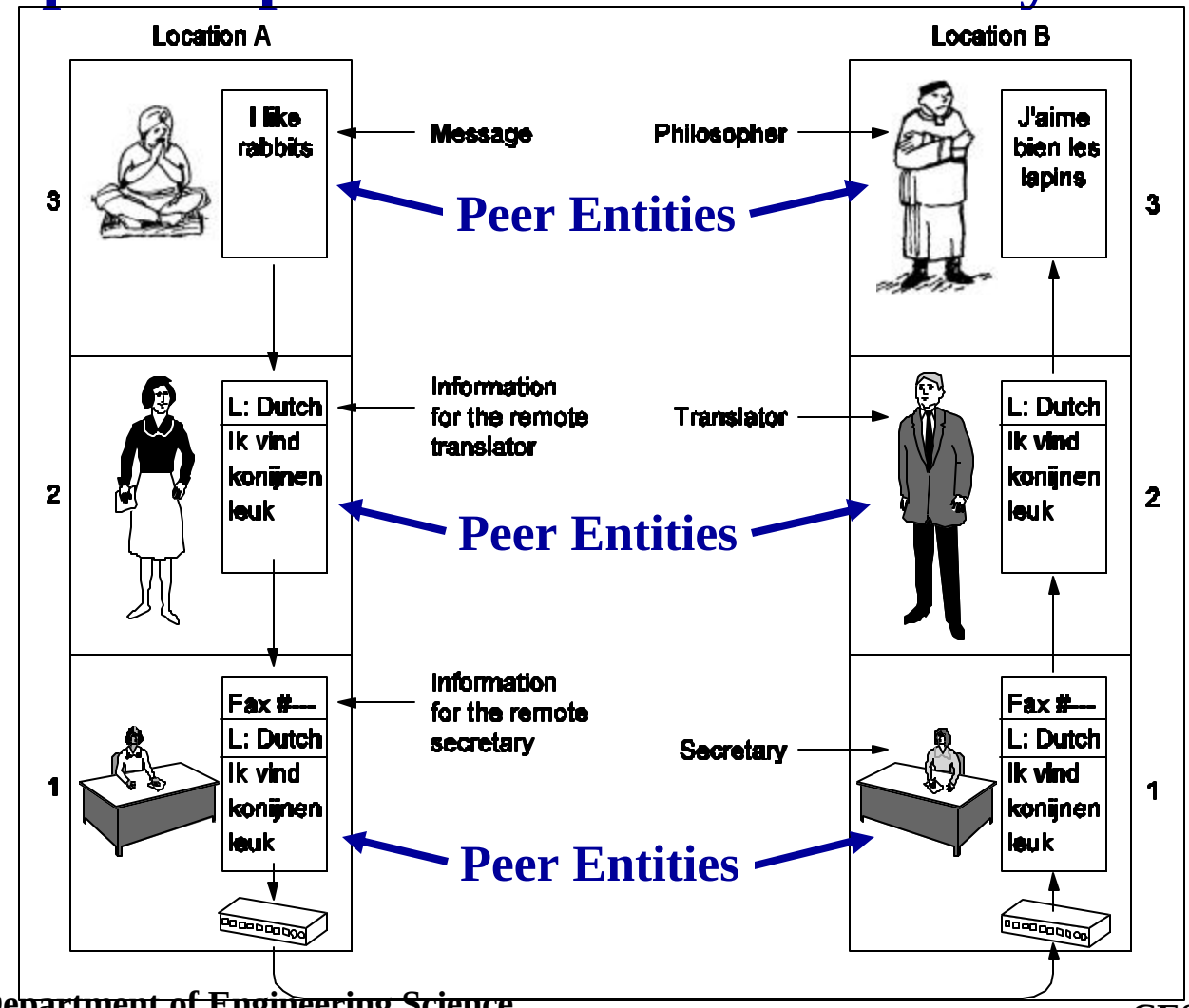
- **Network Architecture: A set of layers and protocols.**

- **Protocol Stack: A list of protocols used by a certain system, one protocol per layer.**

- **Protocol Data Unit (PDU):**
  - A unit of data that is specified by a protocol of a given layer and delivered among peer entities.
  - Consisting protocol control information of given layer and possibly data of that layer.

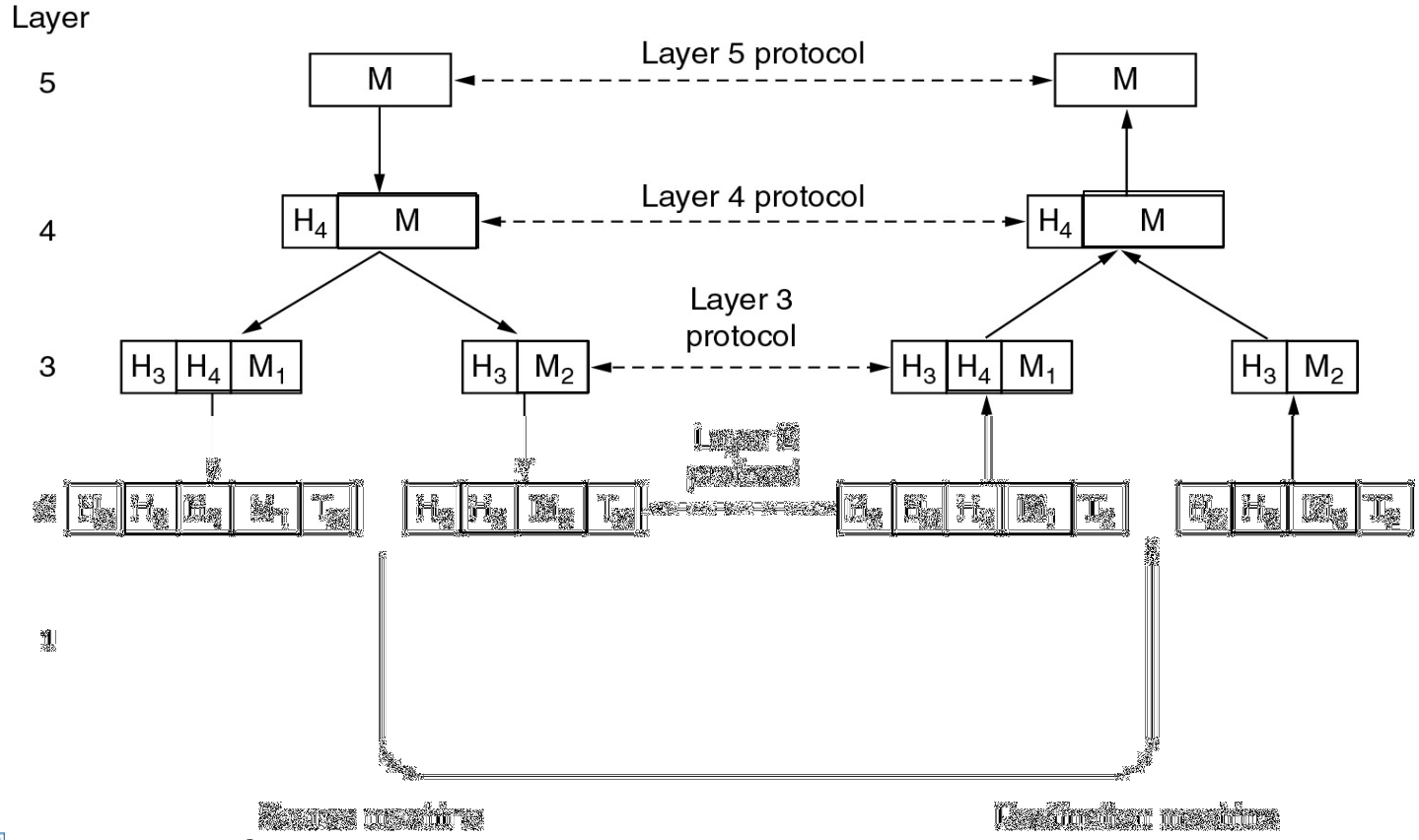
SOFTWARE: PROTOCOL HIERARCHIES

- **The philosopher–translator–secretary architecture**



SOFTWARE: PROTOCOL HIERARCHIES

- **PDU comprises**
  - Control information: How peer entity should handle PDU.
  - Data: Upper layer PDU.



SOFTWARE: DESIGN ISSUES FOR LAYERS

- **Addressing**
  - Identify sender and receiver of a given message.
- **Error Control**
  - Whether the information at the Rx is correct
  - Error Detection, Error Correction.
- **Flow Control**
  - Keep a faster sender from swamping a slow receiver.
- **Multiplexing/Demultiplexing**
  - One low layer connection handles multiple upper layer connections.
- **Segmentation**
  - Some layer cannot handle long messages.
- **Routing**
  - Identifies the path between Tx and Rx.

SOFTWARE: CONNECTION V.S. CONNECIONLESS

- **Connection-Oriented Service**
  - Modeled after telephone system, 3 steps
  - 1. Connection needs to be set up before data transfer.
  - 2. Message is transmitted following the connection.
  - 3. Connection is released after communication is complete.
  - Example: remote login.
- **Connectionless Service (Datagram Service)**
  - Modeled after postal system
  - No connection required.
  - Each packet is routed independently.
  - Order of packet is not guaranteed.
  - Example: e-mail.

SOFTWARE: QUALITY OF SERVICE

- **Each service can be characterized by a quality of service (QoS)**
- **QoS is determined by four elements**
  - Reliability (file transfer)
  - Delay Sensitivity (Internet phone)
  - Jitter Sensitive (video on demand)
  - Bandwidth (multimedia)

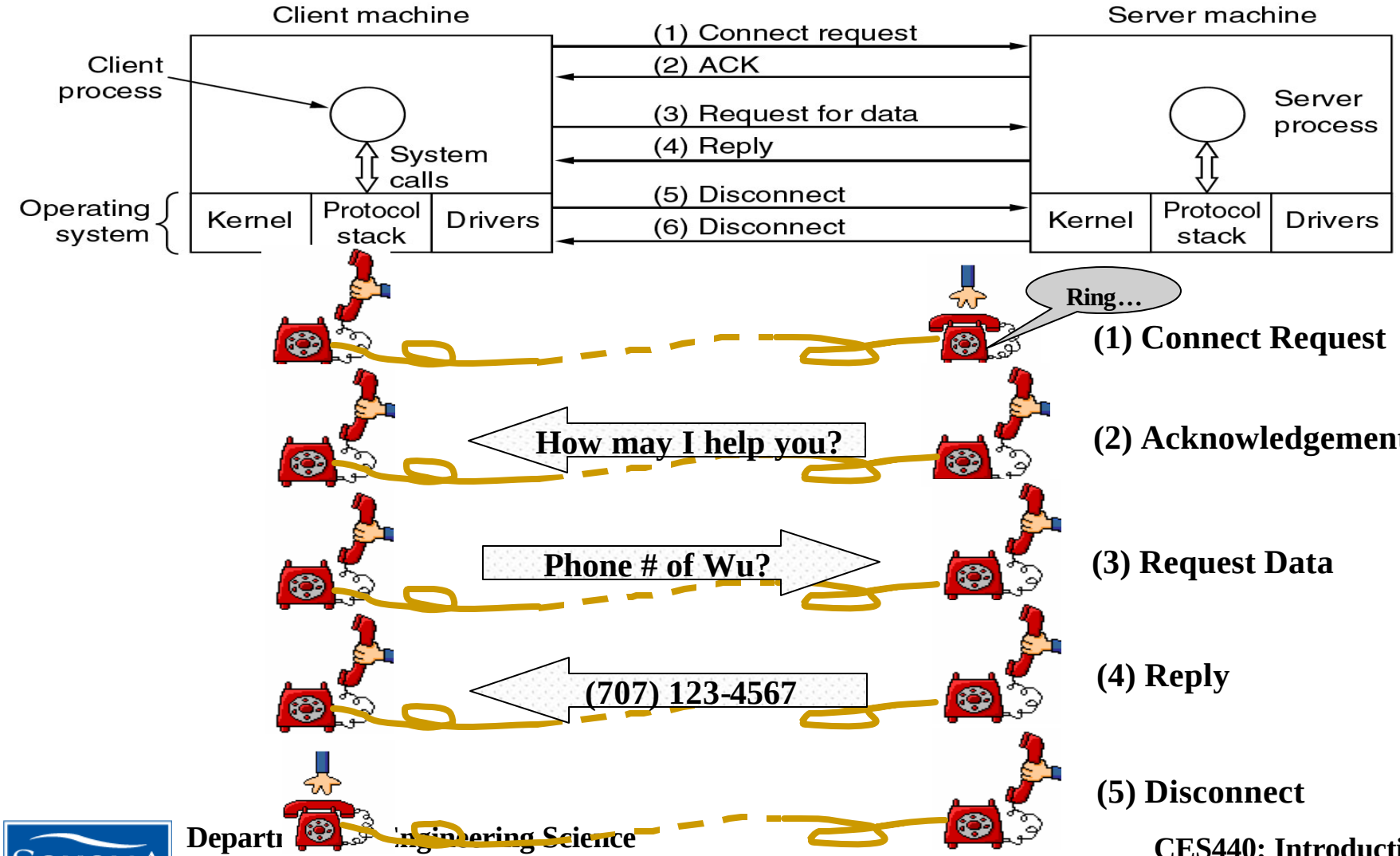
SOFTWARE: SERVICE PRIMITIVES

- **Service Primitives: a set of interface operations available to a user process to access the service.**
  - Usually in the form of a set of functions.
- **An example for a connection oriented service**

Primitives	Meaning
LISTEN	**• Server executes LISTEN – it is prepared to accept a connection** **• Server process is blocked until a request for connection appears**
CONNECT	**• Client process executes CONECT to establish a connection**
RECEIVE	**• Server executes RECEIVE to prepare to accept incoming message**
SEND	**• Client execute SEND to transmit a message**
DISCONNECT	**• Client uses DISCONNECT to terminate the connection**

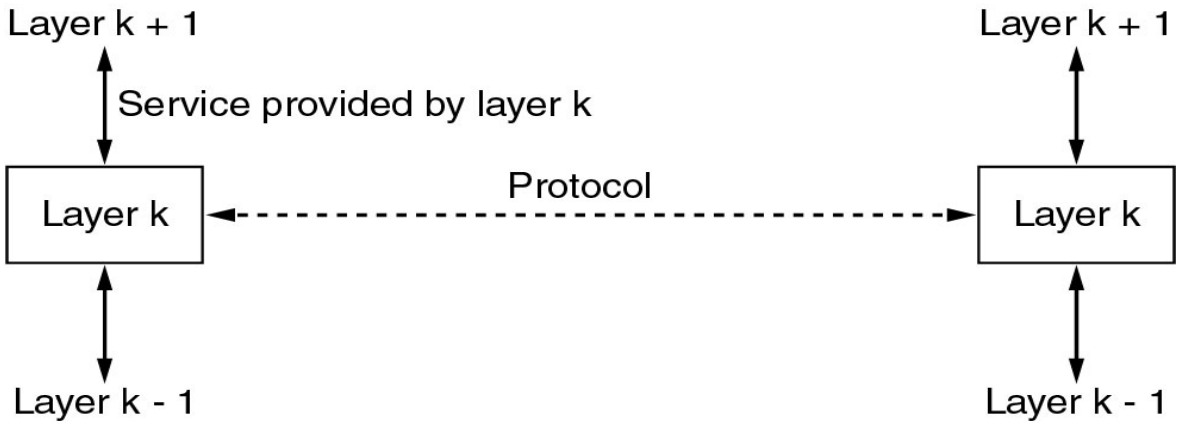
SOFTWARE: SERVICE PRIMITIVES

- **Packets in a simple connection-oriented client-server interaction.**



SOFTWARE: SERVICE V.S. PROTOCOLS

- **Service: A set of primitives (operations) that a layer provides to the layer above it.**
  - Vertical: Interface between two layers.
  - Visible to user (upper layer)
- **Protocol: A set of rules governing the format and handling of packets exchanged by peer entities.**
  - Horizontal: interface between peer entities.
  - Invisible to user (upper layer)



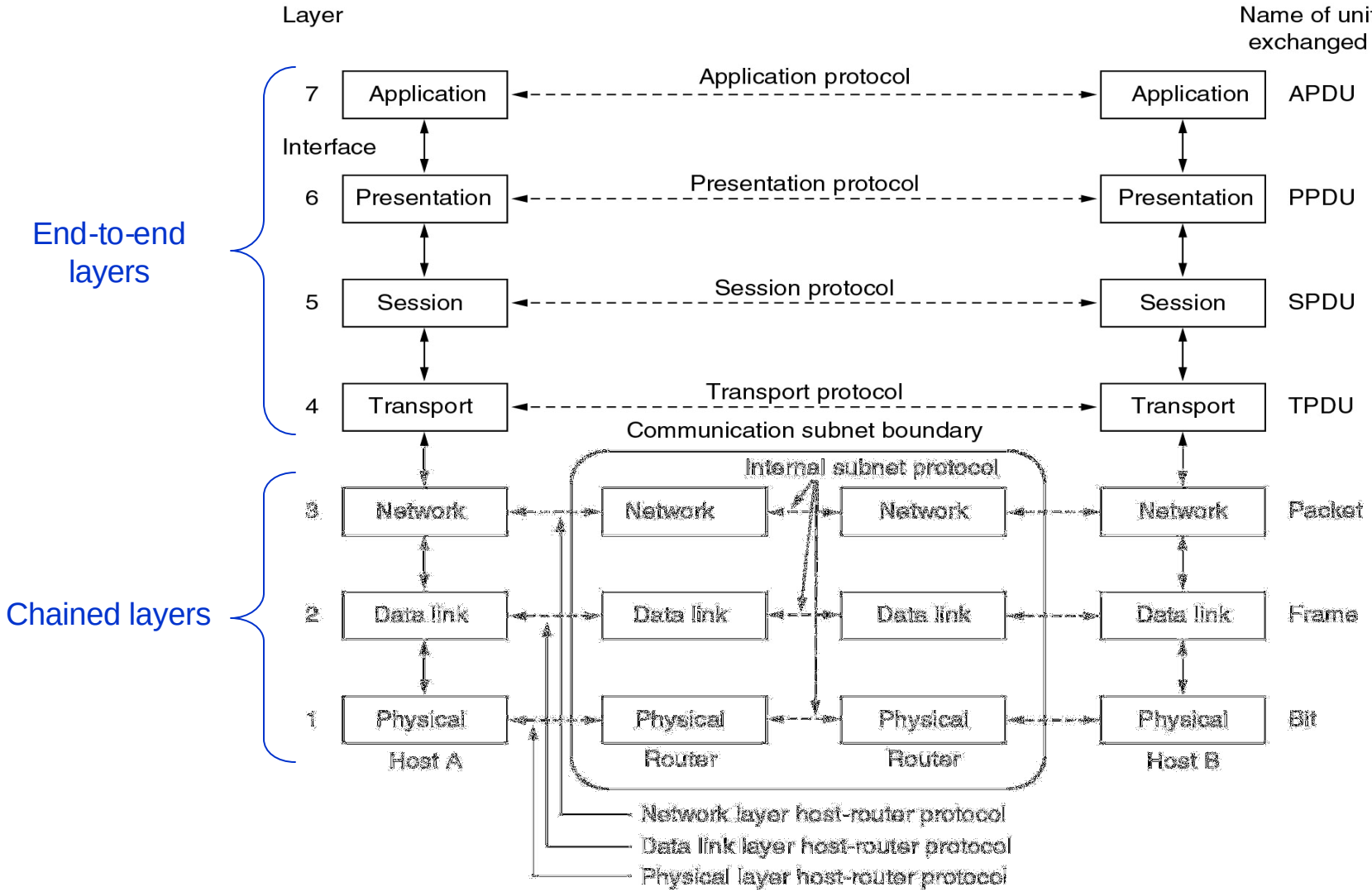
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REFERENCE MODELS: TWO NETWORK STRUCTURES

- **Reference Model**
  - A model defines a certain network structure
    - How are the layers defined? (how many layers? Functions of each layer?)
    - What protocol is used by each layer?
- **The OSI Reference Model**
  - Open System Interconnection
  - Layers are nicely defined with rich features
  - Protocols are rarely used, but is general and valid
- **The TCP/IP Reference Model**
  - Transmission Control Protocol/Internet Protocol
  - Model is not of much use
  - Protocols are widely used
- **We are going to answer the following questions**
  - What are the definitions of the two models?
  - How are the two models compare?
  - Why do we need two models?
  - Which model is better?

REFERENCE MODELS: OSI MODEL



REFERENCE MODEL: OSI MODEL

- **Physical Layer**
  - Concerned with transmitting raw bits over a communication channel
  - The voltage for different symbols (e.g. ‘1’, ‘0’)
  - The duration of each symbol (symbol period)
  - How many pins the network connector has?
  - The function of each network connector pin.
- **Mainly deal with electrical, mechanical, and timing interfaces.**

REFERENCE MODEL: OSI MODEL

- **Data Link Layer**
  - Error Control: make the raw transmission facility (physical layer) appear as error free to the network layer.
    - Error detection, error correction, acknowledgement, retransmission.
  - Flow control: keep a fast transmitter from drowning a slow receiver
    - Traffic regulation
  - Media Access Control: control the access of the shared channel in a broadcast network.
    - Media Access Control (MAC) sublayer

REFERENCE MODEL: OSI MODEL

- **Network Layer: control the operation of subnet**
  - How packets are routed from source to destination.
  - Congestion Control: avoid too many packets swarm the subnet at the same time.
  - Quality of Service (QoS)
  - Allow heterogeneous networks to be interconnected

REFERENCE MODEL: OSI MODEL

- **Transport Layer**
  - Split the data from upper layers into smaller pieces
  - Pass the smaller pieces to lower layer
  - Make sure the pieces arrive correctly at the receiver.
  - Provide an error-free point-to-point channel to upper layers.
- **Transport Layer is true end-to-end**
  - Conversation carried out only between source and destination.
  - Doesn't care the intermediate machines – the intermediate machines doesn't need to pass the message to transport layer.
    - The intermediate machines DO NOT examine the contents of transport header
  - In low layers, conversations are between machines and its immediate neighbors.

REFERENCE MODEL: OSI MODEL

- **Session Layer**
  - Allows users to establish sessions between them
  - Dialog control (whose turn it is to transmit)
  - Token control (preventing conflicts)
  - Synchronization (allow continue from a break point)

- **Presentation Layer**
  - Concerned with syntax and semantics
  - Make it possible for different computers with different data representations (data structures) to talk with each other.

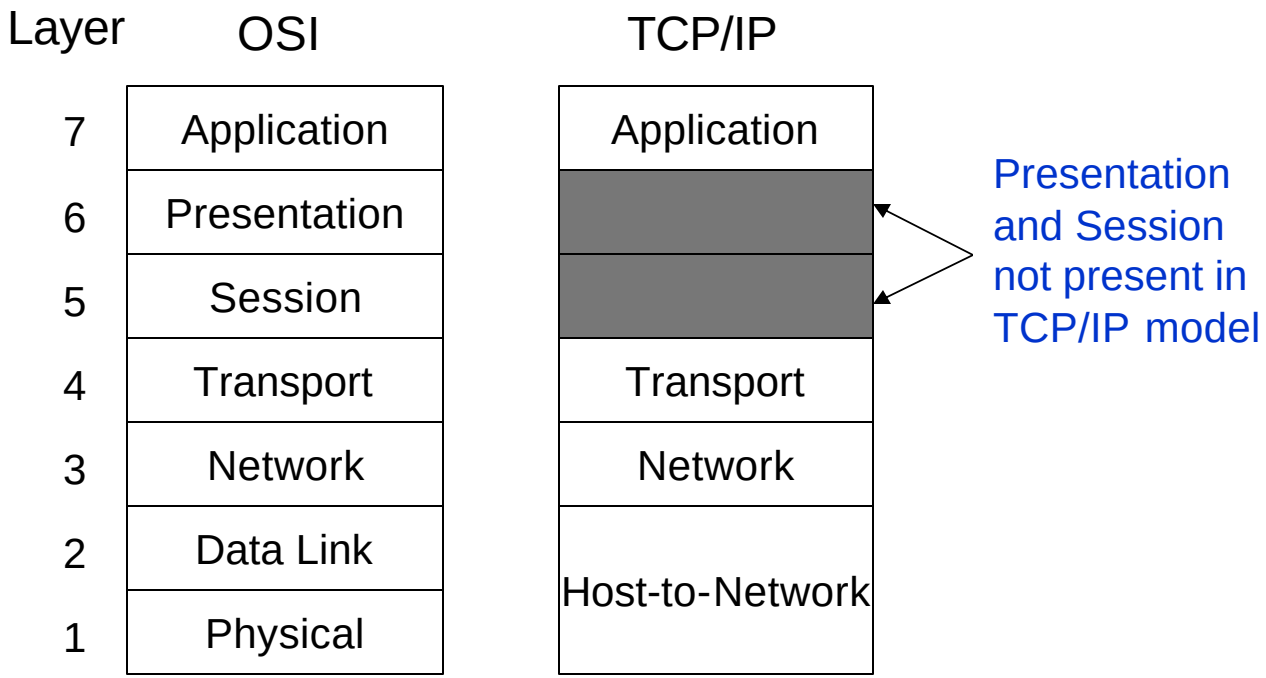
REFERENCE MODEL: OSI MODEL

- **Application Layer**
  - The applications directly used by the end users
  - Examples
    - HTTP (Hyper Text Transfer Protocol)
    - FTP (File Transfer Protocol)
    - E-mail

REFERENCE MODEL: TCP/IP

- **TCP/IP is first used in ARPANET**
  - Advanced Research Project Agency Network
  - ARPANET is a research project supported by DoD (Department of Defense)
  - ARPANET eventually evolves into Internet
  - TCP/IP reference model is developed for ARPANET to make sure the seamless connection of numerous different networks.
- **TCP/IP Model is named after it's two primary protocols**
  - Transmission Control Protocol
  - Internet Protocol

REFERENCE MODEL: TCP/IP



REFERENCE MODEL: TCP/IP

- **Application Layer**
  - All high level protocols on top of transport layer (no session layer and presentation layer)
- **Application Layer Protocols**
  - TELNET (virtual terminal)
  - FTP
  - SMTP (e-mail)
  - DNS (Domain Name Service)
    - Mapping host name to network addresses
  - HTTP
  - …

REFERENCE MODEL: TCP/IP

- **Transport Layer**
  - Allow peer entities to carry on end-to-end conversation (Similar to OSI Model)
- **Transport Layer Protocols**
  - TCP (Transmission Control Protocol)
    - Reliable, Connection-oriented
    - Flow Control
  - UDP (User Datagram Protocol)
    - Unreliable, Connectionless
    - Usually used in applications where delay is more important than reliability

REFERENCE MODEL: TCP/IP

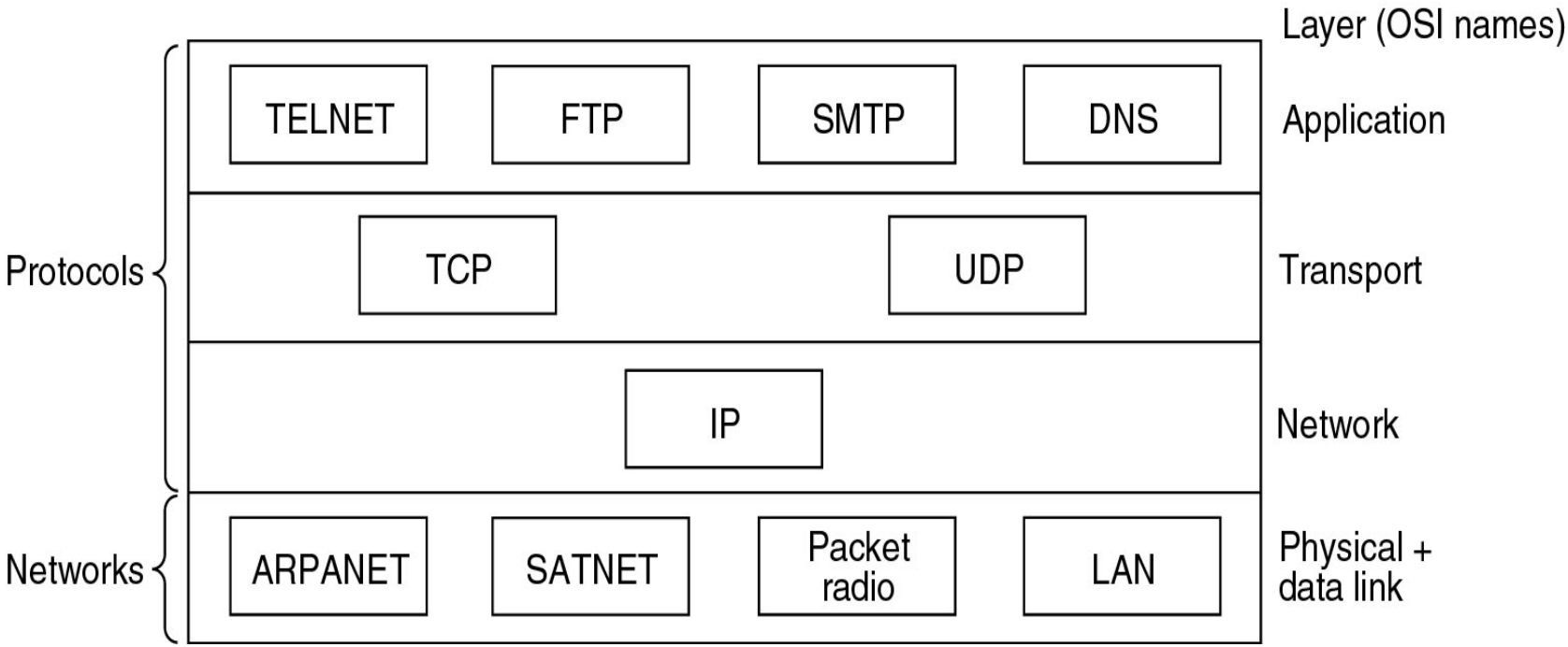
- **Internet Layer: the essence of TCP/IP model.**
  - Connectionless
  - Packet-switching: each packet is transmitted independently
  - Order of packets is not guaranteed
  - Network is able to survive loss of part of the subnet hardware.
  - Analogy: Mail System.
- **Internet Layer Protocol: Internet Protocol (IP)**
  - Packet routing (packet-switching network)
  - Congestion control.
  - Similar to the network layer of OSI model.

REFERENCE MODEL: TCP/IP

- **Host-to-Network Layer**
  - No much details about this layer is given in TCP/IP model.
  - TCP/IP model only specifies that hosts have to connect to the network using some protocol that can send IP packets to it.
  - Protocol is not defined at all.

REFERENCE MODEL: TCP/IP

- **Protocols in the TCP/IP model initially**



REFERENCE MODEL: TCP/IP V.S. OSI

- **Similarities**
  - Both are based on the stack of independent protocols.
  - Functions of layers are similar
    - Layers from bottom up to transport layer provide end-to-end, network independent service to upper layers.
    - Layers above transport layer are application oriented.

REFERENCE MODEL: TCP/IP V.S. OSI

- **Differences**
  - The concept of Services, Interfaces, Protocols
    - Services: What are the layer's functions. Says nothing about how to access it.
    - Interface: How to access the layer. Says nothing about how to implement it.
    - Protocols: How to realize the layers functions (implementation). Can be any as long as the job is done.
  - OSI: makes clear distinction of the 3 concepts
  - TCP/IP: originally didn't distinguish between them.

REFERENCE MODEL: TCP/IP V.S. OSI

- **Differences (Cont'd)**
  - Relation between Protocol and Model
    - OSI: protocols developed after model →protocols are hidden in the model, and can be easily replaced by other protocols as long as the service and interface remain the same.
      - Model are general.
    - TCP/IP: protocols developed before model→model was devised to fit the protocol, and it doesn't fit other protocol stacks.
      - Model are not widely used. Protocols are popular.

REFERENCE MODEL: CRITIQUES OF OSI

- **At late 1980s, many experts thought OSI is the one model that rules all.**
  - It doesn't turn out that way.
- **Why?**
  - Bad timing:when OSI appears, TCP/IP were widely used.
  - Bad Technology.
    - Session and Presentation layers are almost empty.
    - Data Link and Network layer are too crowded.
  - Bad Implementation: too complex for efficient implementation.
  - Bad Politics
    - People thought OSI was pushed by governments.
    - TCP/IP is free and come with UNIX.

REFERENCE MODEL: CRITIQUES OF TCP/IP

- **Distinction between service, interface, and protocol are not well defined.**
  - Not a good software practice.
  - Not much a guide for designing new networks.
- **Not general:**
  - TCP/IP model can only describe TCP/IP protocol.
- **Host-to-network layer is not really a layer**
- **Does not distinguish physical layer and data link layer.**
- **Protocols other than TCP and IP protocols are ad hoc.**
  - Ad hoc: for the particular case at hand without consideration of wider application.
  - Some protocols are deeply entrenched, hard to replace.

REFERENCE MODEL

- **OSI model is exceptionally suitable for discussing the concepts of computer network.**
  - The protocols related to OSI model is obsolete.
- **TCP/IP protocols are widely used.**
  - The model is practically nonexistent.
- **This course uses a hybrid model**

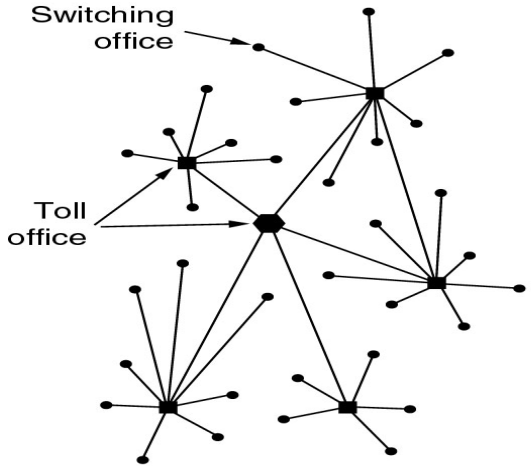
5	Application layer
4	Transport layer
3	Network layer
2	Data link layer
1	Physical layer

OUTLINE

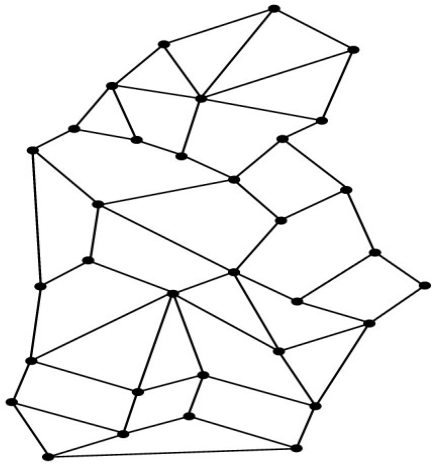
- Computer Network Definition
- Applications of Computer Network
- Network Hardware & Classifications
- Network Software
- Reference Models
- **Example Networks**
- Network Standardizations

EXAMPLE NETWORKS: INTERNET HISTORY

- **ARPANET: Idea**
  - In late 1950s, military comm. use public telephone network
    - Star Topology is vulnerable
  - DoD want a network that can survive nuclear War.
  - Baran proposed distributed switching system
    - Mesh Topology is robust. (AT&T refused to implement it)
  - Advanced Research Projects Agency (ARPA) was founded in 1957
    - Funded BBN to start ARPNET based on the proposal



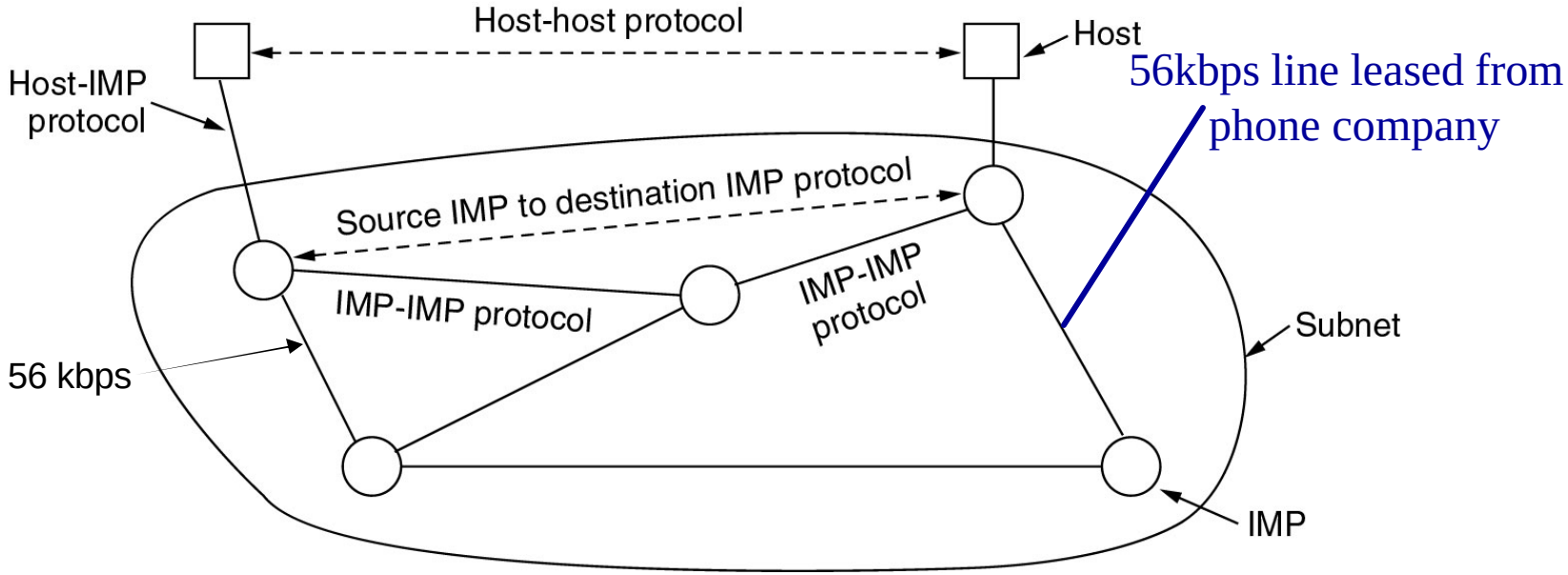
(a)



(b)

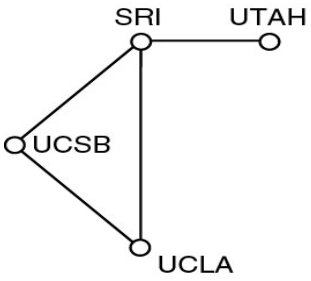
EXAMPLE NETWORKS: INTERNET HISTORY

- **ARPANET: Design**
  - Basically a mesh subnet with IMPs (Interface Message Processors) connecting hosts
  - Multiple paths exist between hosts
  - Using store-and-forward (packet-switching)

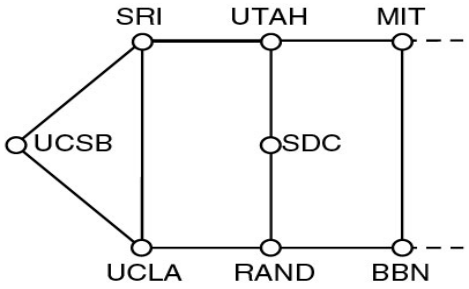


EXAMPLE NETWORKS: INTERNET HISTORY

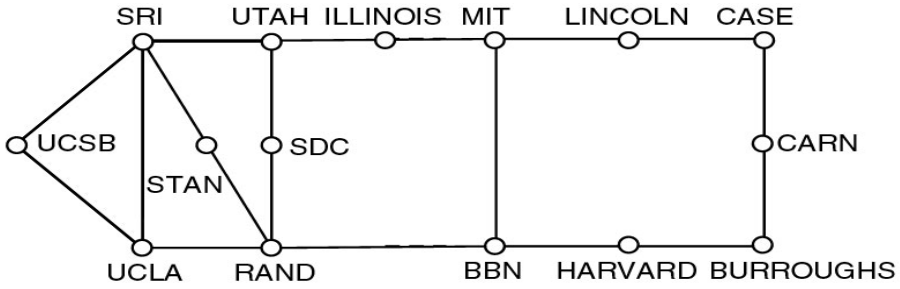
- **ARPNET: Growth**



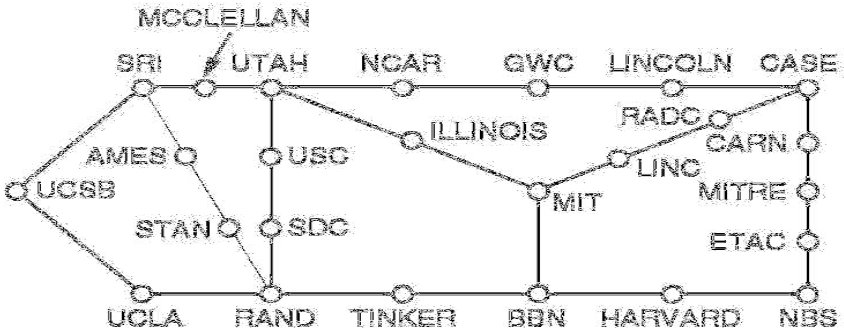
(a) December 1969



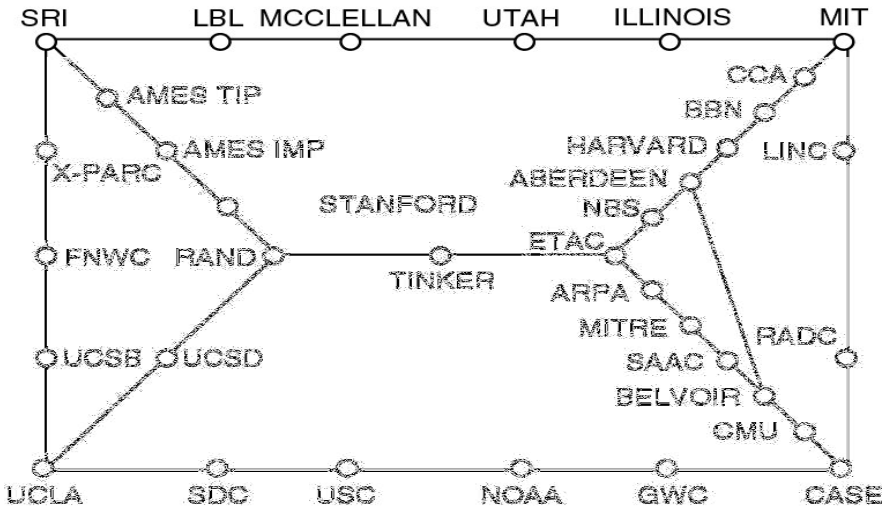
(b) July 1970



(c) March 1971



(d) April 1972



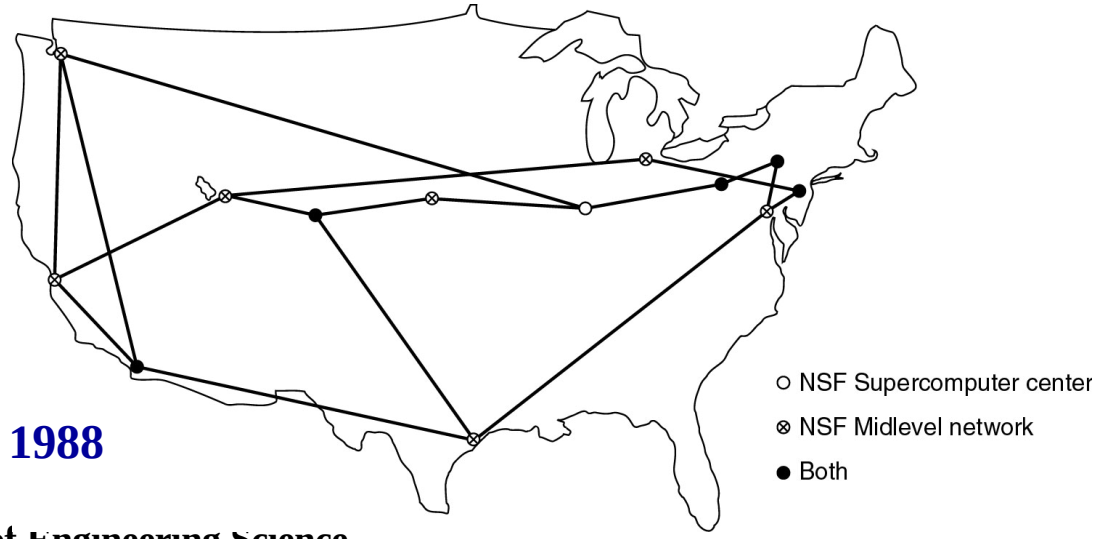
(e) September 1972

EXAMPLE NETWORKS: INTERNET HISTORY

- **ARPNET: Growth (Cont'd)**
  - Original ARPNET protocol was not suitable for internetwork connection
  - TCP/IP was developed to run over internetworks
  - TCP/IP was integrated in Berkeley UNIX
    - With TCP/IP, sockets, network utilities
    - Widely adopted by universities
  - During 1980s, many LANs connected to ARPNET
  - As scale increase, finding hosts become a bottleneck
    - DNS (Domain Name System) was created to map host names onto IP address.

EXAMPLE NETWORKS: INTERNET HISTORY

- **NSFNET**
  - ARPANET can only be connected by universities having research contract with DoD.
  - National Science Foundation (NSF) decided to build a successor of ARPANET can be used by any university.
  - NSF designed a backbone connecting 6 supercomputers and some regional network (1980s)



NSFNET in 1988

EXAMPLE NETWORKS: INTERNET HISTORY

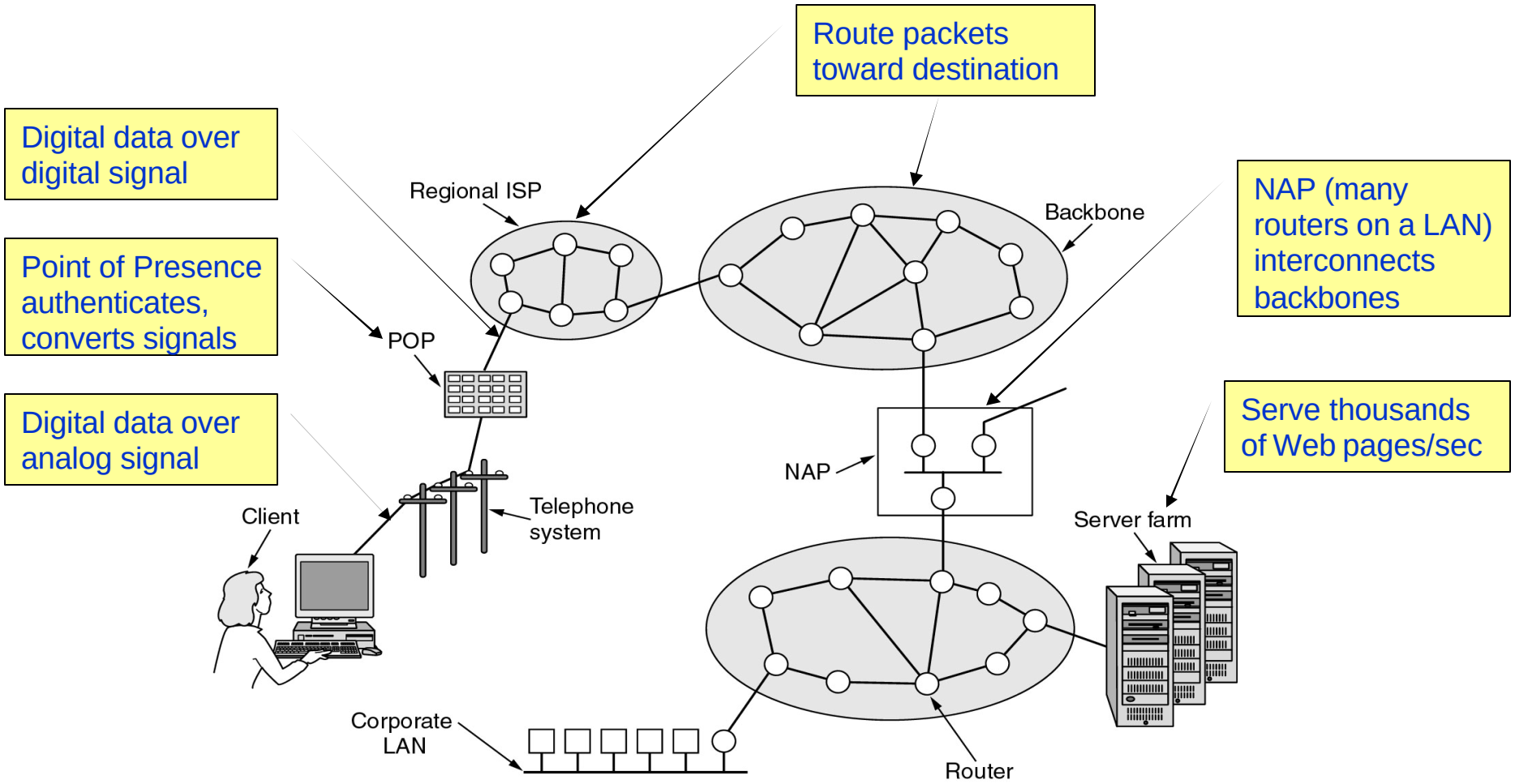
- **NSFNET (Cont'd)**
  - Initially, backbone built with 56kbps leased line
  - Backbone upgraded to 1.5Mbps by 1990.
  - Government could not finance network forever.
  - NSF handed over NSFNET to nonprofit cooperation ANS (Advanced Network and Services)
  - ANS upgrade 1.5Mbps to 45 Mbps to form ANSNET.
  - Big communication companies build their own backbones → competition.
  - Europe developed their own network patterned on ARPNET and NSFNET
  - APRNET and NSFNET are basis of Internet.

EXAMPLE NETWORKS: INTERNET USAGE

- **TCP/IP is the glue of Internet**
  - Make universal service over different networks possible.
  - Analogy: Signaling System 7 for Telephone
- **Traditional Applications (1970 – 1990)**
  - E-mail, News Group, Remote Login, File Transfer
  - Mainly used by academic, government, industry
- **After 1990, new applications were added**
  - WWW (World Wide Web) brings millions of non-academic users
  - E-Commerce, banking, advertising
  - Education
  - Entertainment
  - …

EXAMPLE NETWORKS: INTERNET ARCHITECTURE

- **Client access to server farm via dial-up**

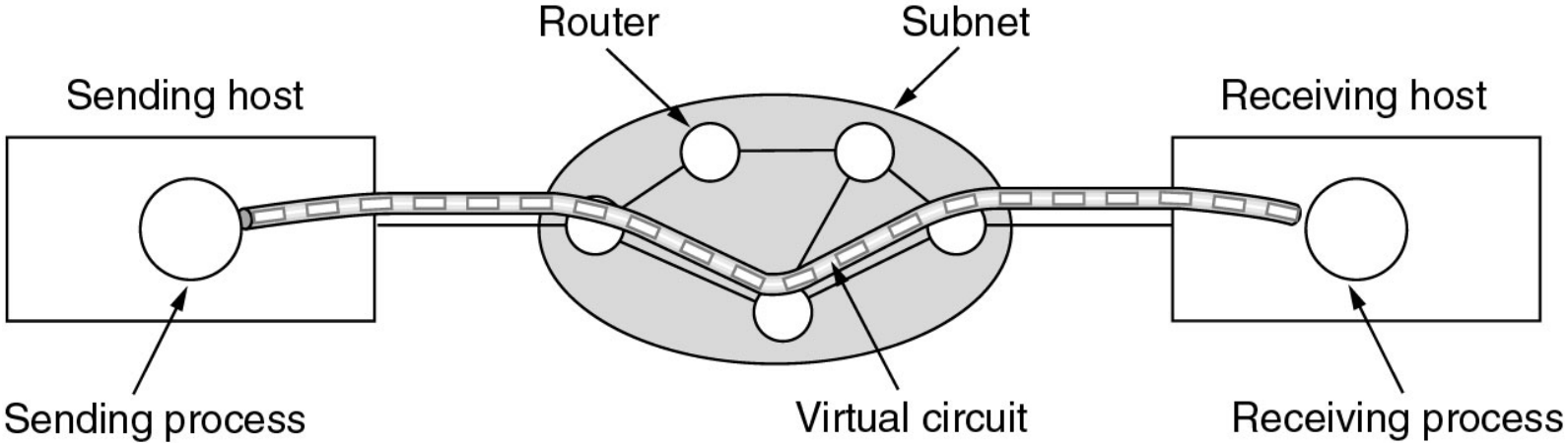


EXAMPLE NETWORKS: CONNECTION ORIENTED NETWORKS

- **Connectionless V.S. Connection Oriented**
  - Connectionless
    - Proponent: ARPANET/Internet community
    - Arguments: faulty tolerant, flexible
  - Connection Oriented
    - Proponent: telephone companies
    - Arguments: QoS, Billing
- **Connection Oriented Examples**
  - X.25: developed in 1970s by phone company for data transmission
  - Frame Relay: 1980s, X.25 replacement, still alive somewhere
    - Interconnecting LANs at multiple company offices.
  - Asynchronous Transfer Mode (ATM): developed in 1990s
    - Widely used within telephone system, and Internet backbones.

EXAMPLE NETWORKS: ATM

- **ATM Virtual Circuits (VC)**
  - During connection setup, each router reserve the necessary resource for the particular connection.
  - Each connection has a connection identifier
  - Permanent VC: permanent connections between two hosts, similar to leased lines.
  - Switched VC: Setup on demand
  - Common Rates: 155Mbps, 622Mbps



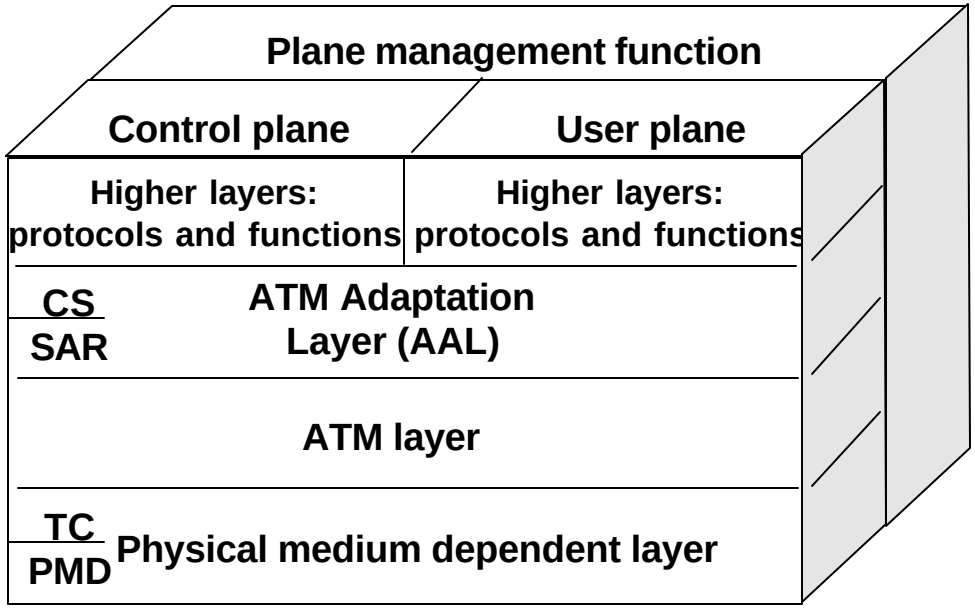
EXAMPLE NETWORKS: ATM

- **ATM Cells**
  - Fixed Length: 53 Bytes
    - Easy to route with hardware (variable length IP packets have to be routed by software)
    - Small size doesn't block the line → Good for QoS
    - Use hardware copy cell to multiple output → Good for HDTV broadcast
  - Header: 5 Bytes
    - Connection Identifier
    - Checksum cover the cell
  - Data: 48 Bytes



EXAMPLE NETWORKS: ATM

- **ATM Reference Model**
  - Three Layers
    - Physical: deals with physical medium, electrical properties
    - ATM: deals with cells and cell transport
    - ATM Adaptation Layer: Segment large packets into cells
  - Three Planes
    - User Plane: user information, error/flow control, segmentation
    - Control plane: signaling, VC setup/release
    - Management Plane: operation, administration, maintenance



CS = Convergence Sublayer
SAR = Segmentation And Reassembly sublayer
TC = Transmission Convergence sublayer
PMD = Physical Medium Dependent sublayer

EXAMPLE NETWORKS: ATM

- **ATM Layers, Sublayers, and their functions**

OSI layer	ATM layer	ATM sublayer	Functionality
3/4	AAL	CS	Providing the standard interface (convergence)
		SAR	Segmentation and reassembly
2/3	ATM		Flow control Cell header generation/extraction Virtual circuit/path management Cell multiplexing/demultiplexing
2	Physical	TC	Cell rate decoupling Header checksum generation and verification Cell generation Packing/unpacking cells from the enclosing envelope Frame generation
1		PMD	Bit timing Physical network access

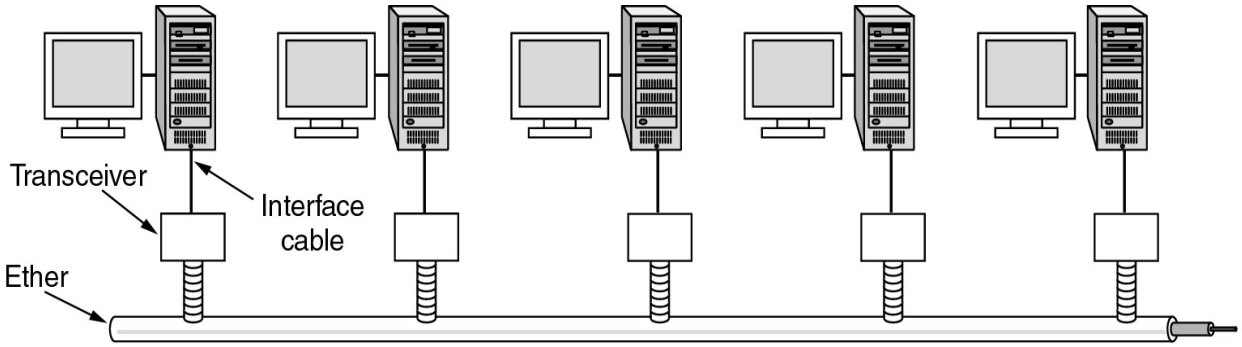
EXAMPLE NETWORKS: ETHERNET

- **ALOHONET – The prototype of Ethernet**
  - University of Hawaii researchers wanted to enable terminals on remote islands to access the main computer in Honolulu.
  - Equip all terminals and main computer with short range radio.
  - Terminals contacting main computer:
    - Terminals send packets at will, and the main computer acknowledge the recipient of the packet.
    - If collision occurs (2 or more terminals Tx simultaneously), no acknowledgement, terminal re-Tx
  - Main computer contacting terminals
    - Only main computer transmits, no collision.

EXAMPLE NETWORKS: ETHERNET

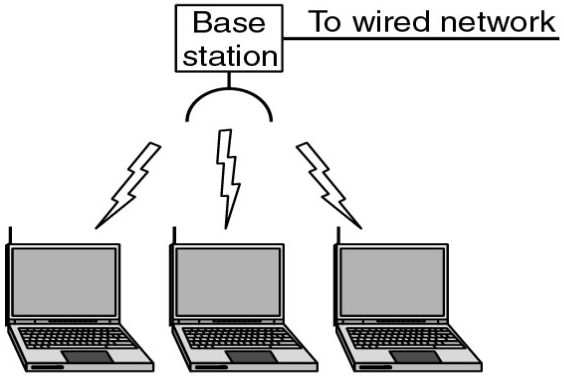
- **Ethernet (IEEE802.3): the most commonly used LAN**
  - All computers share the same cable (ether).
  - Listen to media before Tx
    - If it's idle, transmit.
    - If it's busy, wait until the current Tx is finished.
  - Listen to media during Tx
    - If detect interference (collision), back off, wait a random time before retry.
  - Media: Coaxial Cable (original), twisted pair, fiber
  - Currently 10/100 Mbps, New version 1Gbps.

Original Ethernet

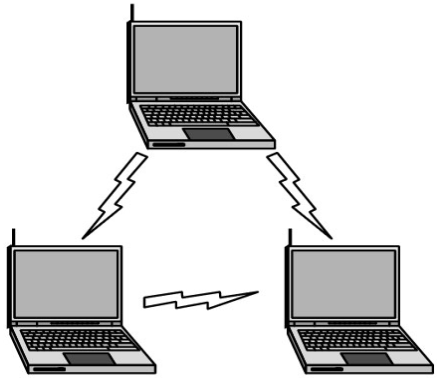


EXAMPLE NETWORKS: WIRELESS LAN

- **Wireless LAN (IEEE 802.11, WiFi)**
  - Provide a short-range mobility and cost effective solution
  - Two modes
    - With base station (access point)
    - Without base station: ad hoc networking
  - Three branches
    - IEEE 802.11a – 54Mbps
    - IEEE 802.11b – 11Mbps
    - IEEE 802.11g – 54Mbps

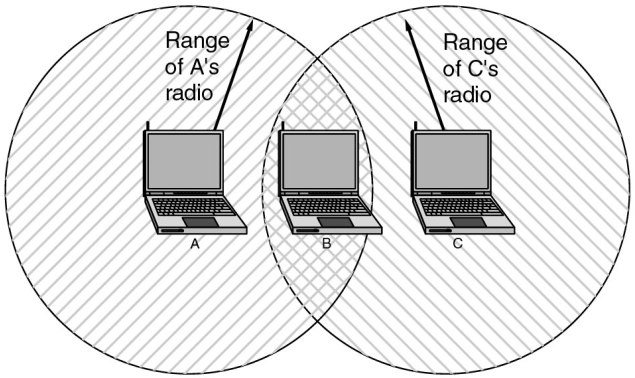


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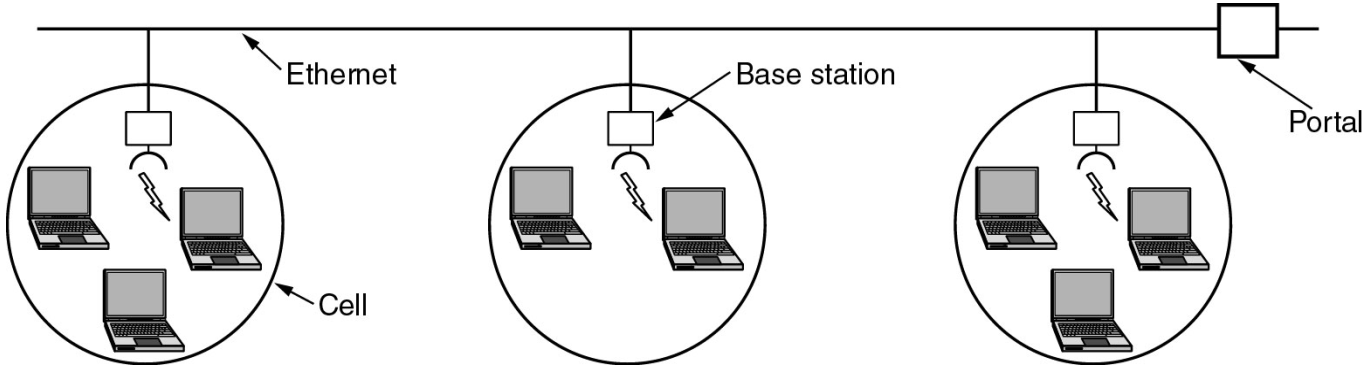

(b)

EXAMPLE NETWORKS: WIRELESS LAN

- **Issues need to be addressed**
  - Cannot listen to media as Ethernet



  - Radios reflection cause amplitude and phase distortion of signals (multipath fading)
  - Mobility and change of base station (hand off in cellular network)



OUTLINE

- Computer Network Definition
- Applications of Computer Network
- Network Hardware & Classifications
- Network Software
- Reference Models
- Example Networks
- **Network Standardizations**

NETWORK STANDARDIZATION

- **Why standardization?**
  - Interoperability
    - Equipments from different manufactures can work together
  - Reduce cost and price
    - Enables large scale production
- **Major standard organizations:**
  - ITU: International Telecommunication Union
    - Telecommunications, V.92 (modem), V.24 (RS-232), X.25, SS7, …
  - ISO: International Standards Organization
    - Computer and other areas, OSI, ISO 9000
  - IAB: Internet Architecture Board
    - Internet, RFC (Request for Comments) (e.g. FTP, Telnet, HTTP)
  - IEEE: Institute of Electrical and Electronics Engineers
    - Networking and other electrical areas, IEEE 802.3, IEEE 802.11

STANDARDIZATION: ITU

- **ITU has three sectors**
  - ITU-R (Radiocommunications)
  - ITU-T (Telecommunications)
  - ITU-D (Development Sector)
- **Structures:**
  - 200 governmental members
    - US: State Department
  - 500 Sector Members: AT&T, Nokia, Compaq, Intel,…
- **Operation:**
  - Make recommendation to governments
  - Government can adopt it or ignore it.

STANDARDIZATION: ISO

- **International Standard Organization**
  - Covers a broad number of subjects
    - Nuts and bolts, fishing nets, computers, etc.
  - In area of telecommunication, often cooperate with ITU-T
    - ISO is a member of ITU-T
  - US representative: ANSI (American National Standards Institute)
  - Standards are designed to achieve as broad a consensus as possible.
  - Example: OSI reference model

STANDARDIZATION: IAB

- **Internet Architecture Board has two branches**
  - IRTF (Internet Research Task Force)
    - Long term research topics
  - IETF (Internet Engineer Task Force)
    - Short term engineering issues
    - Many workgroups working on Internet standards
- **IAB is not as formal as ISO or ITU-T**
- **Procedures**
  - In the early years, RFC (request for comments)
    - www.ietf.org/rfc
  - Later, RFC → Tests → Draft Standard → Internet Standard

STANDARDIZATION: IEEE

- **Institute of Electrical & Electronics Engineers**
  - The largest professional organization in the world.
  - Publishing Journals, running conferences
  - Developing standards in areas of electrical engineering and computing.

Number	Topic
802.1	Overview and architecture of LANs
802.2 ↓	Logical link control
802.3 *	Ethernet
802.4 ↓	Token bus (was briefly used in manufacturing plants)
802.5	Token ring (IBM's entry into the LAN world)
802.6 ↓	Dual queue dual bus (early metropolitan area network)
802.7 ↓	Technical advisory group on broadband technologies
802.8 †	Technical advisory group on fiber optic technologies
802.9 ↓	Isochronous LANs (for real-time applications)
802.10 ↓	Virtual LANs and security
802.11 *	Wireless LANs
802.12 ↓	Demand priority (Hewlett-Packard's AnyLAN)
802.13	Unlucky number. Nobody wanted it
802.14 ↓	Cable modems (defunct: an industry consortium got there first)
802.15 ↑	Personal area networks (Bluetooth)
802.16 ↑	Broadband wireless
802.17	Resilient packet ring

SUMMARY

- **Computer Network Concept**
  - A collection of autonomous computers interconnected with each other
- **Computer Network Applications**
  - Business Applications, home applications, mobile users
  - Broadcast Link, Point-to-point link
- **Hardware and Classifications**
  - PAN, LAN, MAN, WAN, Internetworks
  - Home network, wireless network, etc.
- **Software**
  - Protocol hierarchies, service primitives, layers and protocols, interface v.s. protocol
  - Connection-oriented v.s. connectionless, QoS